Split-Execution of a Heavy Task in a Decentralized System under Non-Uniform Component Distribution and Uniform Event Arrival Rate

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Contents

1 Problem Statement **2**
- 1.1 System Model . . . 2
- 1.2 Non-Uniform Component Distribution . . . 2
- 1.3 Task Arrival Model . . . 2
- 1.4 Sequential Execution Workflow . . . 2
- 1.5 Queueing and Service Dynamics . . . 3

2 Symbols and Their Meaning **3**

3 Modelling Assumptions **4**
- 3.1 Hop-Based Communication Delay Model . . . 4

4 Non-Uniform Geometric Distribution Used **5**

5 Foundational Derivations **5**
- 5.1 M/M/1 Model . . . 5
- 5.2 Derivation of the Standard M/M/1 Formulas . . . 6
- 5.3 Split-Execution Model . . . 6

6 Analysis and Answers **7**
- 6.1 Average Task Completion Time T_{e2e} . . . 7
- 6.2 Average Queue Size Under Each Component . . . 8
- 6.3 Maximum Sustainable Arrival Rate $\lambda_{\max}$ . . . 8
- 6.4 Fraction of Total Delay Due to Queueing and Communication . . . 9
- 6.5 Effect of Non-Uniformity on Delay and Queueing . . . 10

7 Core Formula Sheet **11**

8 Evaluation Results **13**
- 8.1 Observation . . . 13

1 Problem Statement

1.1 System Model

Consider a decentralized mesh network consisting of N interconnected devices $\{D_1, D_2, \ldots, D_N\}$. Any device can route packets to any other via multi-hop forwarding. We examine a computational workload that is decomposed into K ordered components $\{C_1, C_2, \ldots, C_K\}$, where each component represents a functionally distinct sub-task. A complete job requires sequential execution of all components, strictly in the order $C_1 \to C_2 \to \cdots \to C_K$.

1.2 Non-Uniform Component Distribution

Each component C_i is installed on a subset of devices according to a *non-uniform geometric distribution*. Let p_i denote the fraction of devices allocated to host component C_i; equivalently, p_i is the normalized replication share of component C_i. The distribution is defined such that

$$p_1 > p_2 > \cdots > p_K, \qquad \sum_{i=1}^{K} p_i = 1,$$

ensuring that earlier components have broader deployment while later components have progressively narrower presence in the network. This distribution directly controls the *relative computational load* of each component: a smaller p_i implies fewer hosting nodes and thus higher expected queueing and utilization per host under identical demand.

1.3 Task Arrival Model

Tasks originate at random positions in the network. The initiation point of each job is uniformly distributed across all devices, i.e.,

$$\Pr(\text{task originates at } D_j) = \frac{1}{N}.$$

Job arrivals follow a stochastic arrival process (e.g., Poisson), but the uniform spatial distribution ensures no node is preferentially favored for task generation.

1.4 Sequential Execution Workflow

A job begins at its origin device and first requires execution of component C_1. If the origin does not host C_1, the task is forwarded hop-by-hop across the mesh to the nearest (or otherwise selected) device hosting C_1. After completing C_1, the task is forwarded again across the mesh to a device hosting C_2, and so forth, until all K components are completed. Thus the end-to-end execution path of a job is a multi-hop sequence of the form

$$D_{\text{src}} \to D(C_1) \to D(C_2) \to \cdots \to D(C_K).$$

This routing introduces stochastic communication delay that compounds with every component transition.

1.5 Queueing and Service Dynamics

Each device that hosts a component C_i maintains a dedicated queue for incoming requests requiring that component. The arrival rate at a node hosting C_i is affected by:

- the global task arrival rate,
- the fraction p_i of devices hosting C_i,
- the spatial distribution of sources,
- the multi-hop forwarding delays that influence temporal clustering of arrivals.

Because tasks must be processed sequentially, congestion at earlier components can propagate downstream as bursts of completions. Conversely, scarcity of hosting nodes for later components (p_i small) can create bottlenecks due to concentrated load.

2 Symbols and Their Meaning

Symbol	Meaning
N	Total number of devices in the mesh network
K	Number of sequential components required per task
C_i	The i-th component in the processing workflow
p_i	Fraction of devices allocated to component C_i
n_i	Number of hosts for C_i, $n_i = Np_i$
A	Side length of deployment area ($A \times A$)
R	Communication radius of each device
λ	Global task arrival rate (each task uses all components)
$\lambda_i^{(h)}$	Arrival rate to one host of C_i, $\lambda_i^{(h)} = \lambda/(Np_i)$
μ_i	Service rate of a host executing component C_i
ρ_i	Utilization of one host of C_i, $\rho_i = \lambda/(Np_i\mu_i)$
W_i	Sojourn time at component C_i per host (waiting + service)
$W_{q,i}$	Queueing delay at component C_i per host
$L_i^{(h)}$	Average number of tasks at one host of C_i
$L_{q,i}^{(h)}$	Average queue length at one host of C_i
L_i	Total tasks across all hosts of component C_i
$L_{q,i}$	Total queue across all hosts of component C_i
d_0	Average shortest-path hop delay from source to the host of C_1
d_i	Average shortest-path hop delay from a host of C_i to a host of C_{i+1}
T_{mm1}	Sojourn time in one $M/M/1$ queue
T_{e2e}	End-to-end task completion time
$\lambda_{\max}$	Maximum sustainable arrival rate ensuring system stability

3 Modelling Assumptions

- Devices form a connected geometric mesh network in a square area $A \times A$.
- Each device has communication radius R (Random Geometric Graph model).
- Tasks originate uniformly across N devices.
- Component execution order is strictly sequential:

$$C_1 \to C_2 \to \cdots \to C_K.$$

- A device hosting C_i runs an $M/M/1$ queue with service rate μ_i.
- Routing between component hosts is shortest-path multi-hop only; flooding is not considered.
- Inter-component communication delay is modeled using hop count and per-hop delay.
- Load is spread uniformly among all hosts of the same component.

3.1 Hop-Based Communication Delay Model

Let τ_h denote the average delay of one hop. For shortest-path routing, the average hop count between two hosts is approximated by

$$h_i \approx \left\lceil \frac{\bar{\ell}_i}{R} \right\rceil,$$

where $\bar{\ell}_i$ is the expected Euclidean distance between the two nodes. Since the deployment is over a square of side length A, we write

$$\bar{\ell}_i = \alpha_i A,$$

for some geometry factor $0 < \alpha_i < 1$ depending on the source and destination placement statistics. Hence,

$$d_i \approx \left\lceil \frac{\alpha_i A}{R} \right\rceil \tau_h.$$

Similarly,

$$d_0 \approx \left\lceil \frac{\alpha_0 A}{R} \right\rceil \tau_h.$$

4 Non-Uniform Geometric Distribution Used

We adopt a normalized geometric distribution:

$$p_i = \frac{(1-r)r^{\,i-1}}{1-r^K}, \qquad 0 < r < 1,$$

satisfying

$$p_1 > p_2 > \cdots > p_K, \qquad \sum_{i=1}^{K} p_i = 1.$$

5 Foundational Derivations

This subsection records the queueing formulas that are used in the split-execution analysis that follows.

5.1 M/M/1 Model

An $M/M/1$ queue has:

- Poisson arrival rate λ,
- Poisson service rate μ,
- utilization
$$\rho = \frac{\lambda}{\mu},$$
- steady-state probability of having k customers in the system
$$p_k = (1-\rho)\rho^k,$$
- average number of customers in the system
$$L = \sum_{k=0}^{\infty} k\, p_k,$$
- Little's law
$$L = \lambda T_{\mathrm{mm1}},$$
where T_{mm1} is the average time a customer spends in the system.

These are all consistent with the Poisson birth/death process, the $M/M/1$ queue, average queue length, and Little's law.

5.2 Derivation of the Standard M/M/1 Formulas

Using the steady-state distribution,

$$p_k = (1-\rho)\rho^k,$$

the mean number of customers in the system is

$$L = \sum_{k=0}^{\infty} k(1-\rho)\rho^k = \frac{\rho}{1-\rho}.$$

Now apply Little's law:

$$L = \lambda T_{\text{mm1}}.$$

Therefore,

$$T_{\text{mm1}} = \frac{L}{\lambda}.$$

Substituting $L = \rho/(1-\rho)$ and $\rho = \lambda/\mu$ gives

$$T_{\text{mm1}} = \frac{1}{\mu - \lambda}.$$

For a single-server exponential service process with rate μ, the mean service time is

$$\mathbb{E}[S] = \frac{1}{\mu}.$$

Hence the average waiting time in queue is

$$W_q = T_{\text{mm1}} - \frac{1}{\mu}.$$

Also, the average number of customers waiting in queue is

$$L_q = \lambda W_q.$$

Equivalently, using $L = \rho/(1-\rho)$ and the fact that one customer is in service with probability ρ, we get

$$L_q = L - \rho = \frac{\rho^2}{1-\rho}.$$

5.3 Split-Execution Model

For component C_i, let:

$$n_i = Np_i$$

be the number of devices hosting that component.

We assume that the arrival load for component C_i is evenly shared among its n_i hosts.

Therefore, the arrival rate seen by one host of C_i is

$$\lambda_i^{(h)} = \frac{\lambda}{n_i} = \frac{\lambda}{Np_i}.$$

Each host of C_i is then modeled as an $M/M/1$ queue with service rate μ_i. So, by directly substituting

$$\lambda \leftarrow \lambda_i^{(h)}, \qquad \mu \leftarrow \mu_i,$$

we obtain:

$$\rho_i = \frac{\lambda_i^{(h)}}{\mu_i} = \frac{\lambda}{Np_i\mu_i},$$

$$L_i^{(h)} = \frac{\rho_i}{1-\rho_i} = \frac{\lambda_i^{(h)}}{\mu_i - \lambda_i^{(h)}} = \frac{\lambda}{Np_i\mu_i - \lambda},$$

$$W_i = \frac{1}{\mu_i - \lambda_i^{(h)}} = \frac{1}{\mu_i - \lambda/(Np_i)},$$

$$W_{q,i} = W_i - \frac{1}{\mu_i} = \frac{\lambda_i^{(h)}}{\mu_i\left(\mu_i - \lambda_i^{(h)}\right)}.$$

6 Analysis and Answers

6.1 Average Task Completion Time $T_{\mathbf{e2e}}$

For one host of component C_i, the mean time spent in that queueing system is

$$W_i = \frac{1}{\mu_i - \lambda_i^{(h)}} = \frac{1}{\mu_i - \lambda/(Np_i)}.$$

The corresponding mean waiting time in queue is

$$W_{q,i} = W_i - \frac{1}{\mu_i} = \frac{\lambda_i^{(h)}}{\mu_i\left(\mu_i - \lambda_i^{(h)}\right)} = \frac{\lambda/(Np_i)}{\mu_i\left(\mu_i - \lambda/(Np_i)\right)}.$$

Let d_0 denote the average communication delay from the task source to the host of C_1, and let d_i denote the average communication delay from the host of C_i to the host of C_{i+1} for $i = 1, 2, \ldots, K-1$.

Then the average completion time of one task is

$$T_{\text{e2e}} = d_0 + \sum_{i=1}^{K} W_i + \sum_{i=1}^{K-1} d_i.$$

Equivalently,

$$T_{\text{e2e}} = d_0 + \sum_{i=1}^{K} \frac{1}{\mu_i - \lambda/(Np_i)} + \sum_{i=1}^{K-1} d_i.$$

If the total delay is decomposed into queueing, service, and communication parts, then

$$T_{\text{e2e}} = T_{\text{queue}} + T_{\text{service}} + T_{\text{comm}},$$

where

$$T_{\text{queue}} = \sum_{i=1}^{K} W_{q,i}, \qquad T_{\text{service}} = \sum_{i=1}^{K} \frac{1}{\mu_i}, \qquad T_{\text{comm}} = d_0 + \sum_{i=1}^{K-1} d_i.$$

6.2 Average Queue Size Under Each Component

For an $M/M/1$ queue, the average number of tasks in the system at one host of component C_i is

$$L_i^{(h)} = \frac{\rho_i}{1-\rho_i} = \frac{\lambda_i^{(h)}}{\mu_i - \lambda_i^{(h)}} = \frac{\lambda}{Np_i\mu_i - \lambda}.$$

The average number of tasks waiting in the queue at one host is

$$L_{q,i}^{(h)} = L_i^{(h)} - \rho_i = \frac{\rho_i^2}{1-\rho_i} = \frac{\left(\lambda_i^{(h)}\right)^2}{\mu_i\left(\mu_i - \lambda_i^{(h)}\right)}.$$

If the queue size is required for the entire component across all its hosts, then multiplying by the number of hosts $n_i = Np_i$ gives

$$L_i = n_i L_i^{(h)} = Np_i \cdot \frac{\lambda}{Np_i\mu_i - \lambda}.$$

Similarly, the total queue length across all hosts of C_i is

$$L_{q,i} = n_i L_{q,i}^{(h)}.$$

6.3 Maximum Sustainable Arrival Rate $\lambda_{\max}$

For stability of the queue at component C_i, we must have

$$\lambda_i^{(h)} < \mu_i.$$

Substituting $\lambda_i^{(h)} = \lambda/(Np_i)$ gives

$$\frac{\lambda}{Np_i} < \mu_i \qquad \Longrightarrow \qquad \lambda < Np_i\mu_i.$$

Since this must hold for every component, the maximum sustainable arrival rate is

$$\lambda_{\max} = \min_{1\le i\le K} \left(Np_i\mu_i\right).$$

If all components have the same service rate $\mu_i = \mu$, then

$$\lambda_{\max} = N\mu \min_i p_i = N\mu p_K,$$

because p_K is the smallest fraction under the non-uniform geometric placement.

6.4 Fraction of Total Delay Due to Queueing and Communication

The total queueing delay accumulated over all components is

$$T_{\text{queue}} = \sum_{i=1}^{K} W_{q,i}.$$

The total communication delay is

$$T_{\text{comm}} = d_0 + \sum_{i=1}^{K-1} d_i.$$

The total service time is

$$T_{\text{service}} = \sum_{i=1}^{K} \frac{1}{\mu_i}.$$

Hence,

$$T_{\text{e2e}} = T_{\text{queue}} + T_{\text{service}} + T_{\text{comm}}.$$

Therefore, the fraction of total delay due to queueing is

$$F_{\text{queue}} = \frac{T_{\text{queue}}}{T_{\text{e2e}}},$$

and the fraction due to communication is

$$F_{\text{comm}} = \frac{T_{\text{comm}}}{T_{\text{e2e}}}.$$

If the service fraction is also needed, it is

$$F_{\text{service}} = \frac{T_{\text{service}}}{T_{\text{e2e}}},$$

with

$$F_{\text{queue}} + F_{\text{comm}} + F_{\text{service}} = 1.$$

6.5 Effect of Non-Uniformity on Delay and Queueing

Because the placement is non-uniform and geometric, the fractions satisfy

$$p_1 > p_2 > \cdots > p_K.$$

Since the per-host arrival rate is

$$\lambda_i^{(h)} = \frac{\lambda}{Np_i},$$

a smaller p_i implies a larger arrival rate per host. Consequently, the utilization

$$\rho_i = \frac{\lambda_i^{(h)}}{\mu_i} = \frac{\lambda}{Np_i\mu_i}$$

increases as p_i decreases.

This directly increases queueing delay, because

$$W_i = \frac{1}{\mu_i - \lambda/(Np_i)}$$

and

$$L_i^{(h)} = \frac{\lambda}{Np_i\mu_i - \lambda}.$$

Hence, as the component placement becomes more concentrated, the later components become more heavily loaded, their queues grow faster, and the bottleneck shifts toward the least replicated component. In particular:

- smaller $p_i \Rightarrow$ larger per-host arrival rate,
- larger per-host arrival rate $\Rightarrow$ larger utilization,
- larger utilization $\Rightarrow$ larger queue length and waiting time,
- the component with the smallest p_i typically determines $\lambda_{\max}$.

Thus, non-uniformity increases both delay and queueing imbalance across the pipeline, and the overall system performance is dominated by the least replicated component.

7 Core Formula Sheet

1. M/M/1 Queue

$$\rho = \frac{\lambda}{\mu}$$

$$p_k = (1-\rho)\rho^k$$

$$L = \frac{\rho}{1-\rho}$$

$$T = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}$$

$$\mathbb{E}[S] = \frac{1}{\mu}$$

$$W_q = T - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu-\lambda)}$$

$$L_q = L - \rho = \frac{\rho^2}{1-\rho}$$

2. Component Placement and Load Splitting

$$n_i = Np_i$$

$$\lambda_i^{(h)} = \frac{\lambda}{n_i} = \frac{\lambda}{Np_i}$$

$$\rho_i = \frac{\lambda_i^{(h)}}{\mu_i} = \frac{\lambda}{Np_i\mu_i}$$

3. Per-Host Queueing Quantities for Component C_i

$$W_i = \frac{1}{\mu_i - \lambda_i^{(h)}} = \frac{1}{\mu_i - \lambda/(Np_i)}$$

$$W_{q,i} = W_i - \frac{1}{\mu_i} = \frac{\lambda_i^{(h)}}{\mu_i(\mu_i - \lambda_i^{(h)})}$$

$$L_i^{(h)} = \frac{\rho_i}{1-\rho_i} = \frac{\lambda}{Np_i\mu_i - \lambda}$$

$$L_{q,i}^{(h)} = \frac{\rho_i^2}{1-\rho_i} = \frac{(\lambda/(Np_i))^2}{\mu_i(\mu_i - \lambda/(Np_i))}$$

4. Component-Wide Queue Lengths

$$L_i = n_i L_i^{(h)}$$

$$L_{q,i} = n_i L_{q,i}^{(h)}$$

5. End-to-End Task Completion Time

$$T = d_0 + \sum_{i=1}^{K} W_i + \sum_{i=1}^{K-1} d_i$$

$$T_{\text{queue}} = \sum_{i=1}^{K} W_{q,i}$$

$$T_{\text{service}} = \sum_{i=1}^{K} \frac{1}{\mu_i}$$

$$T_{\text{comm}} = d_0 + \sum_{i=1}^{K-1} d_i$$

$$T = T_{\text{queue}} + T_{\text{service}} + T_{\text{comm}}$$

6. Delay Fractions

$$F_{\text{queue}} = \frac{T_{\text{queue}}}{T}$$

$$F_{\text{comm}} = \frac{T_{\text{comm}}}{T}$$

$$F_{\text{service}} = \frac{T_{\text{service}}}{T}$$

$$F_{\text{queue}} + F_{\text{comm}} + F_{\text{service}} = 1$$

7. Maximum Sustainable Arrival Rate

$$\lambda_i^{(h)} < \mu_i \quad \Rightarrow \quad \lambda < N p_i \mu_i$$

$$\boxed{\lambda_{\max} = \min_{1 \le i \le K} (N p_i \mu_i)}$$

If $\mu_i = \mu$:

$$\lambda_{\max} = N \mu \, p_K$$

8. Non-Uniform Geometric Placement

$$p_i = \frac{(1-r)r^{i-1}}{1-r^K}, \qquad p_1 > p_2 > \cdots > p_K, \qquad \sum_{i=1}^{K} p_i = 1.$$

8 Evaluation Results

To validate the analytical expressions, a discrete-event simulation was implemented in Python. The simulation follows the same modeling assumptions used in the derivation: Poisson arrivals, $M/M/1$ service at each component host, uniform load splitting across replicas, and shortest-path communication delay represented by fixed average hop delays.

The evaluation uses the following sample parameters:

$$N = 100, \quad K = 5, \quad r = 0.75, \quad \lambda = 8, \quad \mu_i = 1.2\ \forall i, \quad d_0 = 1, \quad d_i = 1\ \forall i.$$

The corresponding component replication counts obtained from the geometric placement are:

$$(n_1, n_2, n_3, n_4, n_5) = (33, 25, 18, 14, 10).$$

The comparison between theoretical and simulation values is shown in Table 1.

Table 1: Comparison of theoretical and simulated results

Component	Hosts	Theory W_i	Sim W_i	Theory $W_{q,i}$	Sim $W_{q,i}$	Theory L_i	Sim L_i
C_1	33	1.0443	1.0450	0.2110	0.2115	0.2532	0.2529
C_2	25	1.1364	1.1369	0.3030	0.3070	0.3636	0.3638
C_3	18	1.3235	1.3084	0.4902	0.4814	0.5882	0.5815
C_4	14	1.5909	1.5864	0.7576	0.7531	0.9091	0.8970
C_5	10	2.5000	2.4603	1.6667	1.6320	2.0000	1.9682

The end-to-end delay is:

$$T_{\text{e2e}}^{\text{theory}} = 12.5951, \qquad T_{\text{e2e}}^{\text{sim}} = 12.5370.$$

The simulation closely matches the theoretical values. The small mismatch is due to finite sample size and stochastic variability in arrivals and service times. The error remains low across all components, which validates the analytical queueing model.

8.1 Observation

As p_i decreases, the number of hosts for component C_i decreases, the per-host load increases, and both queueing delay and average queue length increase. The last component, having the smallest replication share, behaves as the bottleneck of the pipeline.